

Belief Heterogeneity and Information Acquisition

When Others Make the Decision*

Dario Trujano-Ochoa[†]

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Abstract

When should you pay for information that someone else will act on it? Standard Bayesian theory predicts it should not matter, and this paper tests that prediction experimentally.

In a laboratory experiment, pairs face the Bayesian decision problem of purchasing a signal about a binary state. I independently vary (i) whether participants share common or heterogeneous prior beliefs when eliciting their posteriors, and (ii) whether the information purchaser is also the final decision maker. I elicit first- and second-order posterior beliefs via the strategy method and willingness to pay (WTP) for information using a BDM mechanism.

Three main results emerge. First, participants expect others to update beliefs similarly to themselves. Second, WTP is significantly higher when participants implement their own strategy under heterogeneous priors, consistent with an amplified control premium when belief differences are salient. Third, individual deviations from Bayesian updating predict WTP in theoretically consistent directions.

These findings suggest that belief heterogeneity can increase the perceived value of decision rights. This has implications for organizational information acquisition and delegation design.

JEL Classification: D83, D91, C91

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[†]Texas A&M University. dario.trujanochoa@tamu.edu